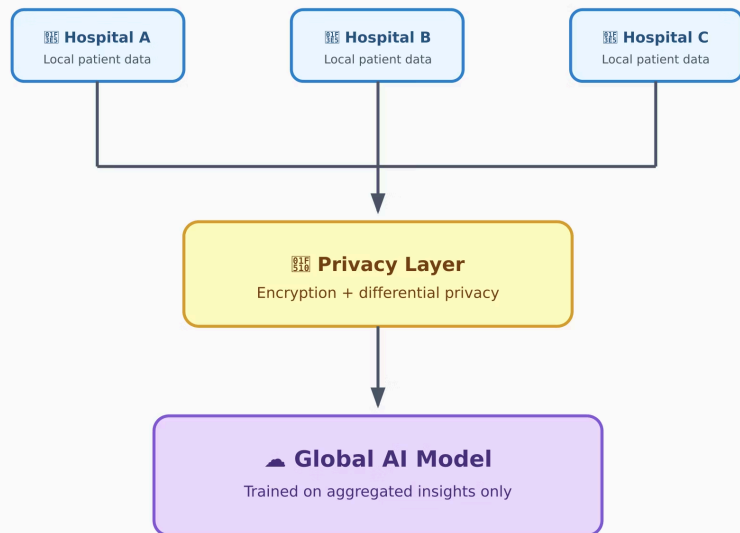


Federated Learning Architecture



Raw patient data never leaves the hospital —
only encrypted model updates are shared

Differential Privacy Trade-offs in Federated Learning for Healthcare Diagnostics

Privacy-Preserving Collaborative AI for Medical Diagnosis

Can hospitals collaboratively train an AI diagnostic model without sharing patient data?

Our project combines **Federated Learning + Differential Privacy** to allow hospitals to collaboratively train medical AI while keeping patient data local.

Problem Statement

Current healthcare AI development faces a fundamental conflict:

 AI models need more data, but healthcare organizations cannot freely share patient data.

Key Problems:

Data Privacy

Medical images may contain personally identifiable and sensitive information.

Centralized Data Risk

Collecting patient datasets in one location creates a high-value target for data breaches.

Data Silos

Hospitals independently possess valuable datasets that cannot easily be combined.

Problem Definition:

How can hospitals collaboratively train accurate diagnostic AI models while minimizing privacy risks associated with both patient data and shared model updates?

Motivation

? Why this project?



HealthCare Need

More diverse data can improve AI diagnostic performance. Individual hospitals may not have sufficient data.



Privacy Need

Medical information is highly sensitive. Institutions need privacy-preserving collaboration.



AI Need

Federated Learning enables distributed model training. Differential Privacy strengthens protection against information leakage.



Research Need

Privacy comes at a potential cost to model performance. The optimal balance is not simply "maximum privacy".

Existing System

Centralized Machine Learning

Hospitals transfer medical datasets to a centralized server.

Advantages:

- Easy model training
- High data availability
- Straightforward implementation

Limitations:

- Raw patient data must leave hospitals
- Higher privacy and security risk
- Centralized point of failure
- Difficult to obtain data from multiple institutions

Federated Learning

Hospitals keep their data locally and exchange model updates.

Advantages:

- Raw data remains local
- Collaborative training
- Reduced need for centralized datasets

Limitations:

- Model updates may still contain information that could potentially reveal properties of training data.

Approach	Raw Data Sharing	Collaboration	Privacy
Centralized ML	Yes	High	Low
Federated Learning	No	High	Better
FL + DP	No	High	Stronger

Research Gap

Although Federated Learning reduces the need to share raw medical data, **privacy risks can remain in shared model updates**.

Differential Privacy can reduce these risks, but introducing noise can reduce model utility.

Existing Gap

Most implementations focus primarily on achieving either:

High model performance

Strong privacy

rather than systematically studying their practical trade-off.

Research Gap Diagram

Differential Privacy

Adds protection but introduces noise.

Privacy vs Utility

Noise can reduce model performance.

Federated Learning

Keeps data local but updates may leak.

Our Research

Measures the trade-off empirically.

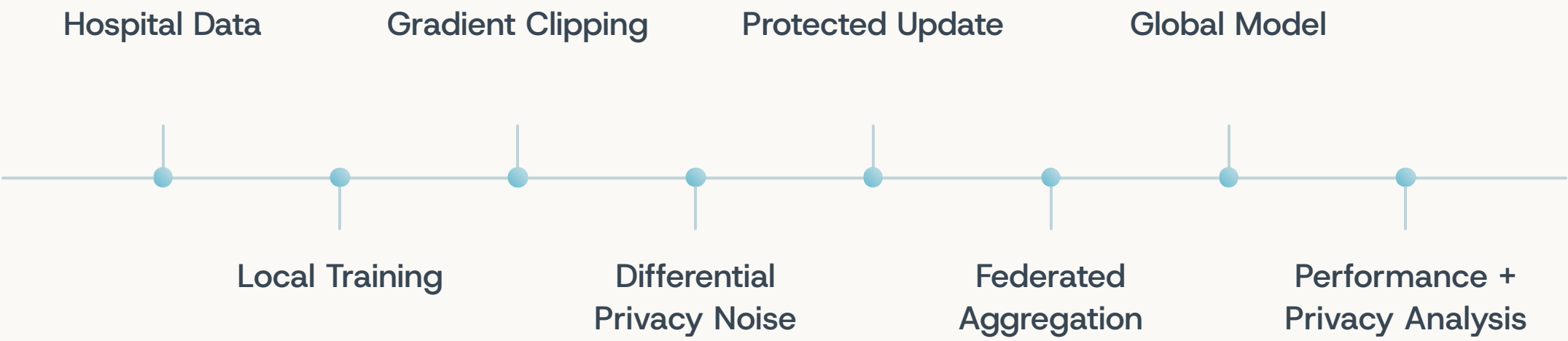
This is your most important research slide.

Proposed System

We propose a **Privacy-Preserving Federated Healthcare AI Framework** combining:

- Federated Learning
- Differential Privacy
- Federated Aggregation
- Research Layer

Core Pipeline



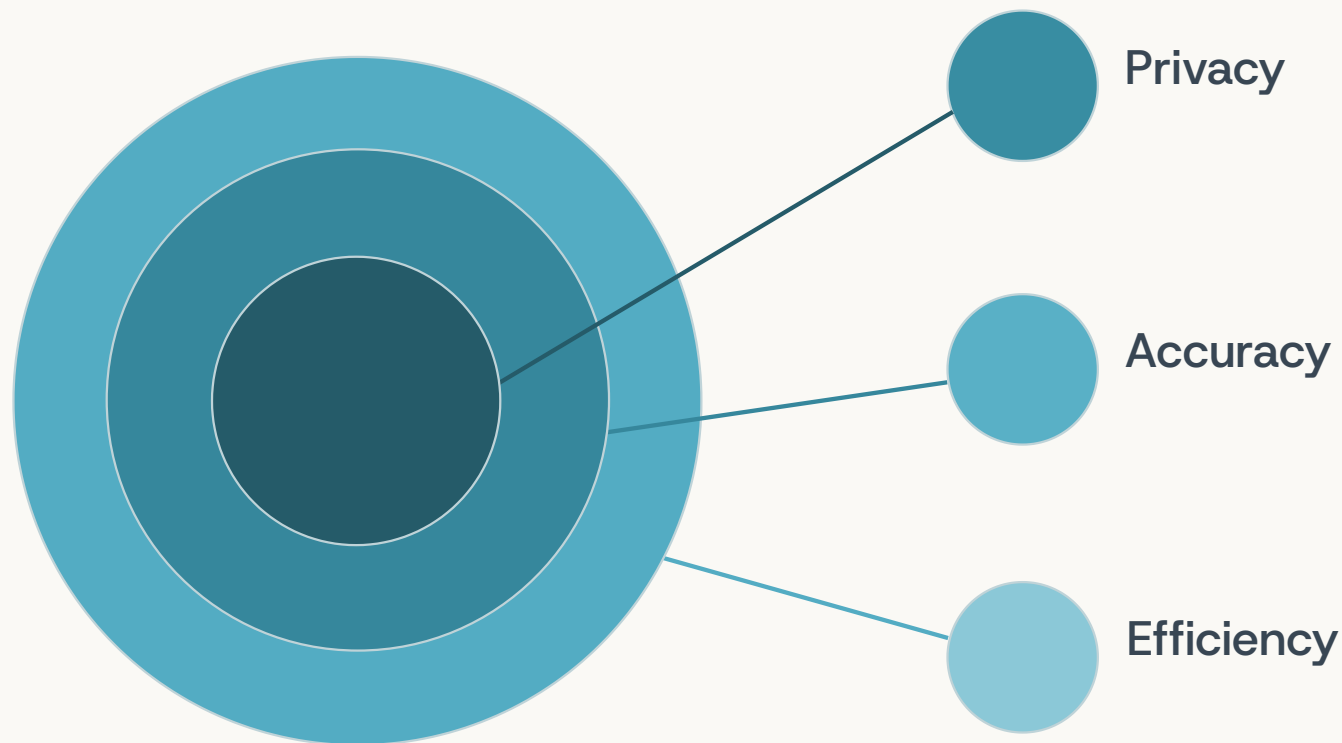
Objectives

Primary Objectives

- Develop a Federated Learning framework for medical image classification.
- Keep raw patient data within individual hospital clients.
- Integrate Differential Privacy into the federated training process.
- Quantify the impact of different privacy levels on diagnostic performance.
- Analyze privacy–utility trade-offs under IID and non-IID data distributions.
- Compare centralized learning, Federated Learning, and FL with Differential Privacy.

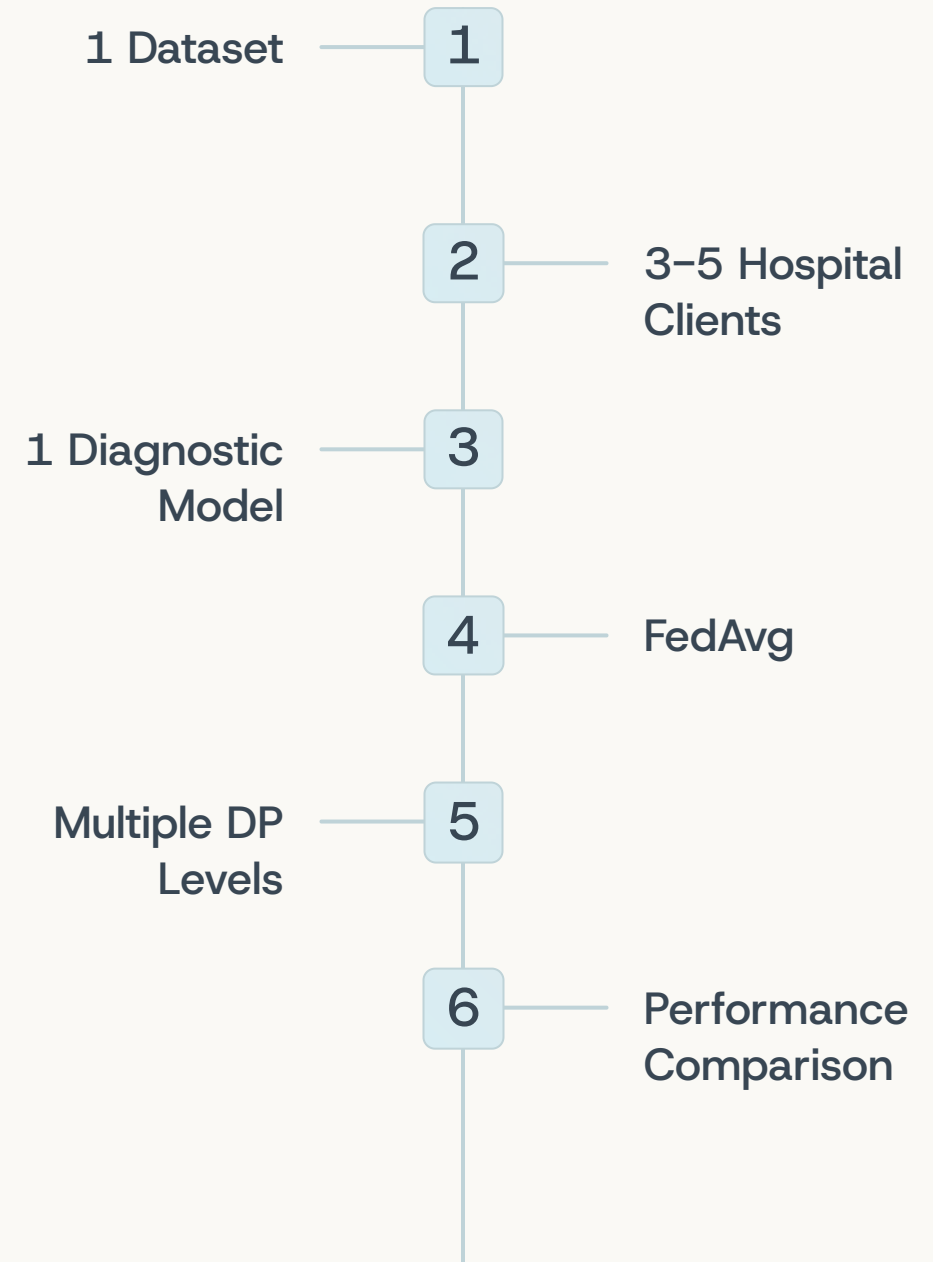
Secondary Objectives

- Analyze model convergence across federated rounds.
- Measure computational and communication overhead.
- Develop a dashboard for experiment monitoring and visualization.
- Identify a practical privacy configuration that maintains acceptable diagnostic performance.



Project Scope

- Publicly available medical imaging dataset
- Multiple simulated hospital clients
- Deep-learning-based disease classification
- Federated Learning using FedAvg
- Differential Privacy
- IID and non-IID data experiments
- Multiple privacy budgets/noise levels
- Performance evaluation
- Privacy–utility analysis
- Visualization dashboard



Methodology

1

Dataset Preparation

Select and preprocess a public medical imaging dataset.

2

Hospital Simulation

Partition the dataset into multiple virtual hospitals. Both **IID and non-IID distributions** will be considered.

3

Centralized Baseline

Train the model centrally to establish a performance baseline.

4

Federated Training

Each hospital trains the global model locally.

5

Differential Privacy

Gradient Clipping → Noise Addition → Protected Update

6

Federated Aggregation

Server combines client updates using **FedAvg**.

7

Evaluation

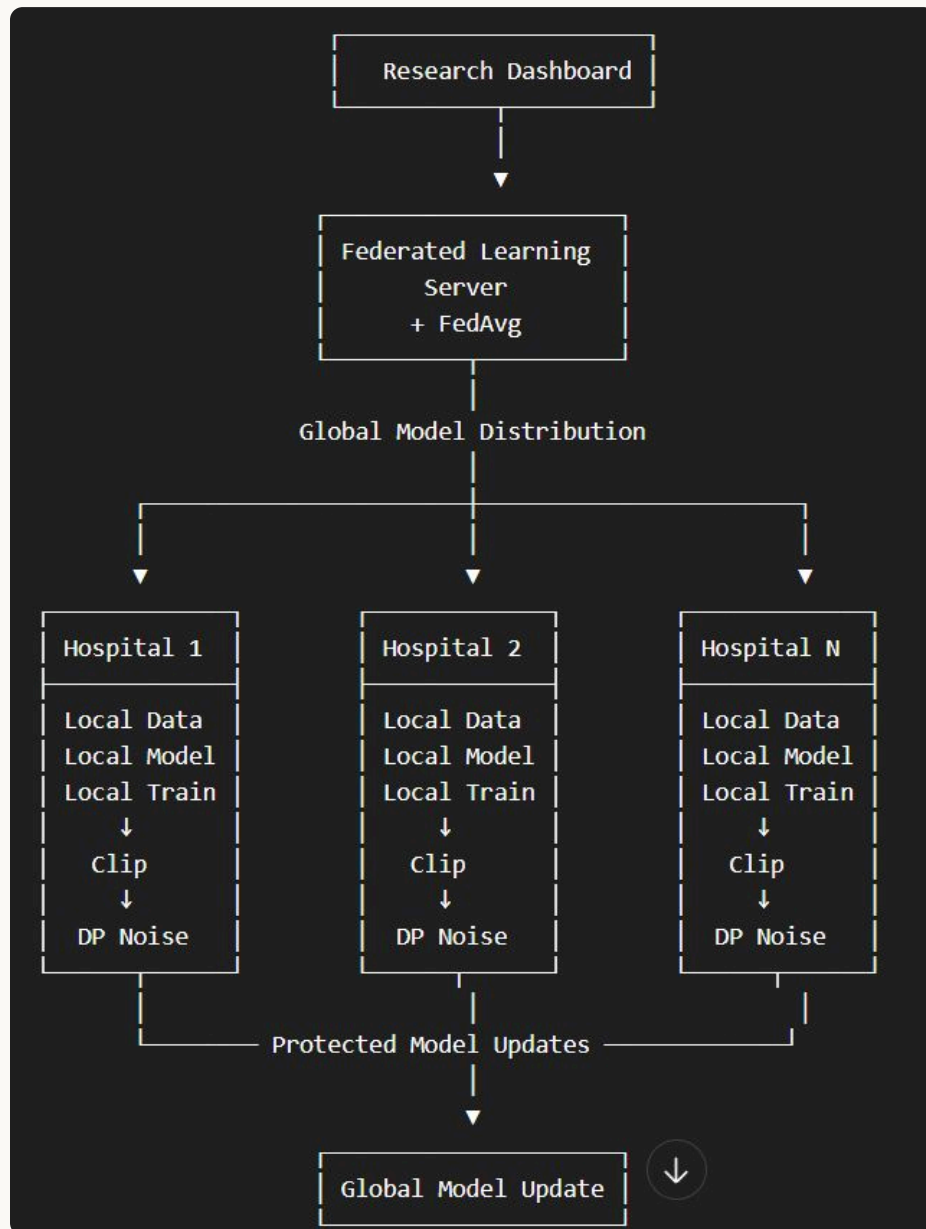
Measure: Accuracy, Precision, Recall, F-1 Score, ROC-AUC Curve, Loss, Convergence

8

Trade-off Analysis

Compare diagnostic performance against privacy strength.

System Architecture



🔒 Raw patient images never leave the hospital client.

Technologies/ Tools Used

Machine Learning

- Python
- PyTorch
- Scikit-learn
- NumPy
- Pandas

Federated Learning

- Flower
- FedAvg

Differential Privacy

- Opacus
- Gaussian Differential Privacy
- Gradient Clipping

Dataset/ Medical Imaging

- Public medical imaging dataset
- OpenCV / PIL
- Image preprocessing and augmentation

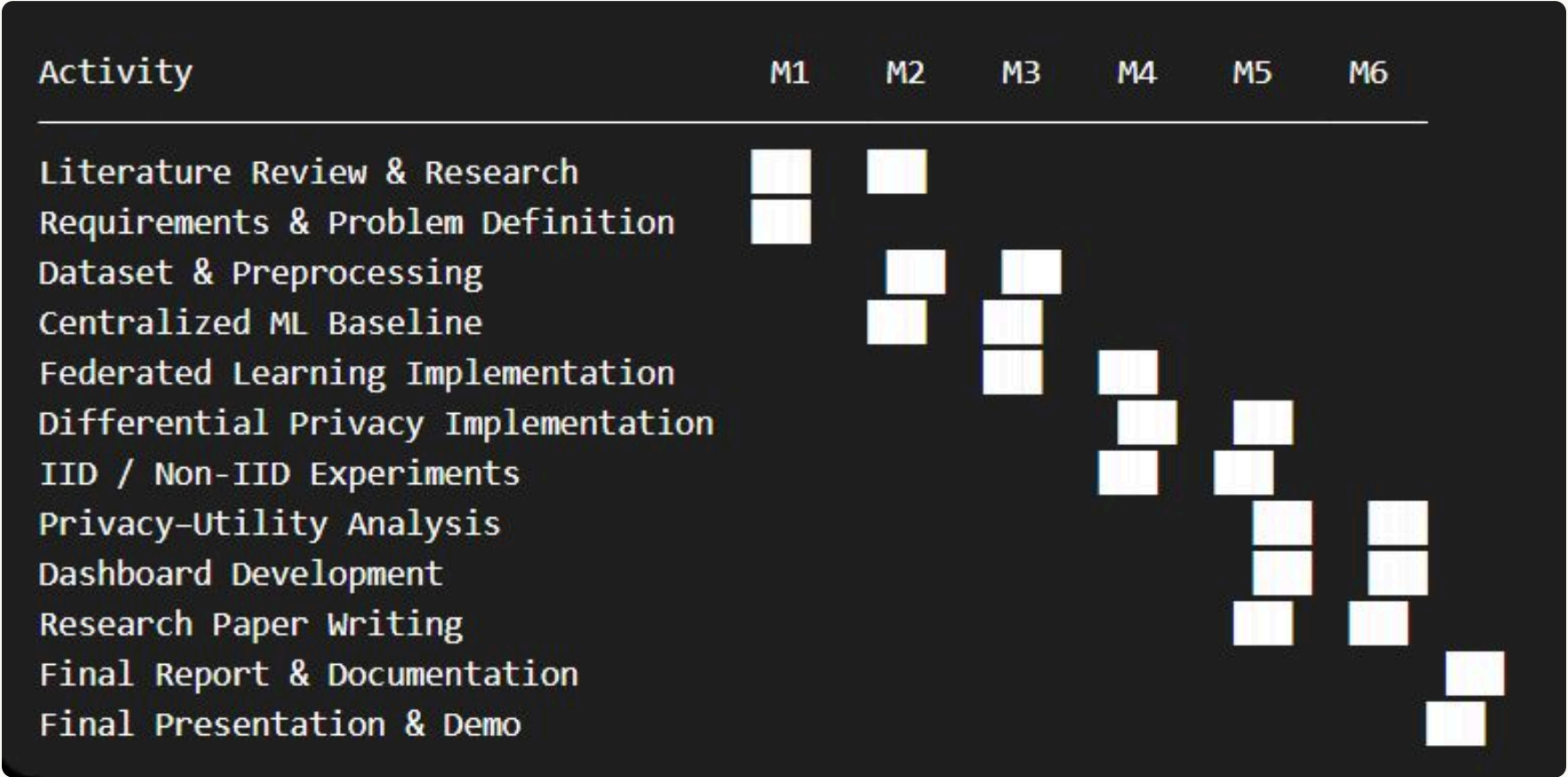
Dashboard/ Application

- Streamlit
- Python-based experiment dashboard

Development

- Jupyter / Google Colab
- Git & GitHub
- VS Code

Project Timeline/ Gantt Chart



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